

From Overlays to Operators: Relational Spatial Weights for Data-Driven Policy

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Abstract

Data-driven urban policy increasingly relies on spatial econometric models and machine learning systems to analyze complex spatial processes. However, most spatial analytical frameworks continue to rely on simple proximity-based spatial weights matrices derived from geographic adjacency or distance decay. These assumptions often fail to capture the relational structure of spatial systems, where interactions are mediated by zoning regimes, infrastructure networks, institutional boundaries, and environmental systems.

This note introduces *Relational Spatial Weights (RelWeights)*, a framework that transforms thematic spatial overlays into formal spatial operators usable within spatial econometrics and geospatial data science. RelWeights allow the construction of irregular spatial lags, enable empirical identification of layer-specific spatial externalities, and provide a mechanism for transferring spatial information across geographic scales.

Conceptually, RelWeights bridge the longstanding divide between the geometrical conception of *space* common in regional science and the contextual conception of *place* emphasized in planning and urban theory. By translating qualitative overlay relationships into quantitative spatial operators, RelWeights provide a framework for modeling spatial dependence in complex, layered spatial systems encountered in real-world policy analysis.

Space, Place, and the Limits of Conventional Spatial Weights

Spatial econometrics and regional science have historically modeled spatial dependence through geometrically defined spatial weights matrices based on contiguity, distance, or nearest-neighbor relationships (Anselin, 1988). These approaches reflect the assumption that spatial interaction primarily operates through geographic proximity.

However, urban systems frequently exhibit relational dependencies that cannot be adequately captured through purely geometric representations of space. Urban theory has long emphasized the importance of *place* as a contextual construct shaped by institutional, infrastructural, and socioeconomic relationships (Portugali, 2011). In planning practice, these contextual relationships are frequently analyzed through overlay techniques that combine multiple spatial layers such as zoning, infrastructure networks, environmental systems, and administrative boundaries.

Despite their conceptual importance, such overlay relationships have rarely been incorporated directly into spatial econometric models. As a result, many spatial models rely on weights matrices that may fail to capture the actual mechanisms generating spatial dependence in real-world spatial systems.

Recent advances in machine learning such as (Agarwal et al., 2024; Brown et al., 2025) have further expanded the analytical possibilities for spatial data through the development of high-dimensional spatial embeddings derived from satellite imagery, mobility data, and large-scale socioeconomic datasets. These embeddings represent spatial units as vectors in a latent feature

space, capturing complex contextual relationships that are difficult to encode through traditional variables. However, a persistent challenge in applying such representations within spatial analysis is that embeddings are often generated at spatial supports that do not align with the units used in policy analysis or econometric modeling. As a result, there is a growing need for formal operators capable of transferring or propagating these representations across heterogeneous spatial supports. Relational spatial weights provide one such mechanism: by defining spatial similarity through shared contextual overlays rather than purely geometric adjacency, they allow embeddings learned at one spatial layer to be systematically transferred or diffused to another.

Relational Spatial Weights

Relational Spatial Weights (RelWeights) extend the conventional concept of spatial contiguity by allowing spatial relationships to be inherited across overlapping spatial layers. Initially proposed in Khedgikar (2023), the idea builds on two important observations discussed in Anselin (1988). First, even within regular lattice structures, the definition of contiguity is not unique. Different criteria such as Queen, Rook, or Bishop contiguity represent different interpretations of spatial interaction.

Second, as emphasized in other works such as Anselin (1980, 1982, 1984), the specification of the spatial weights matrix should ideally reflect the underlying spatial process being studied. The choice of spatial weights should therefore be informed by the contextual structure of the spatial system.

RelWeights operationalize these insights by allowing spatial relationships to be inherited through overlay structures. Consider two spatial layers with the same geographic extent:

$$\alpha = \{1, 2, \dots, n\}, \quad \beta = \{1, 2, \dots, k\}$$

where layer α represents the primary spatial units of analysis and layer β represents an overlaid spatial structure such as zoning regimes, land use classification units, resource catchments, or infrastructure networks.

Let

$$W_{n \times k}$$

denote the intersection matrix between the two layers, where each element W_{ij} represents the degree of intersection between spatial unit i in layer α and spatial unit j in layer β .

Each row vector

$$w_i = (W_{i1}, W_{i2}, \dots, W_{ik})$$

describes the contextual footprint of spatial unit i across the overlay layer.

From this intersection matrix we construct the relational weights matrix

$$R = WW' - \text{diag}(WW')$$

The matrix WW' is a Gramian matrix measuring similarity between contextual footprints of spatial units. Specifically,

$$(WW')_{ij} = \sum_{l=1}^k W_{il}W_{jl}$$

captures the extent to which spatial units i and j share intersections with the overlay layer.

Subtracting the diagonal removes self-similarity and produces a relational weights matrix describing inherited spatial relationships across the overlay structure.

This formulation allows spatial units to become connected even when they are not directly contiguous in the original spatial lattice. Spatial dependence can therefore propagate along contextual structures embedded in the landscape rather than strictly through geometric adjacency.

Example: RelWeights in Data-Driven Spatial Analysis

To illustrate the construction of RelWeights, consider a simple two-layer spatial system. Let the primary layer consist of four districts

$$\alpha = \{A, B, C, D\}$$

and let the overlay layer consist of four contextual regimes

$$\beta = \{X, Y, Z, W\}.$$

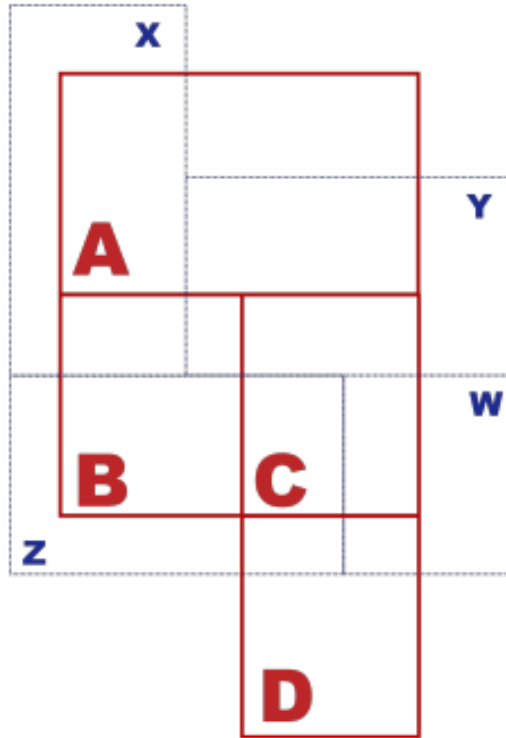


Figure 1: Intersection of α and β

Suppose the intersection relationships between the two layers are given as follows:

- A intersects X, Y
- B intersects X, Y, Z
- C intersects W, Y, Z
- D intersects W, Z

These relationships can be explained visually as shown in Figure 1, or through an intersection matrix as follows:

$$W = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

where rows correspond to districts and columns correspond to contextual regimes (X, Y, Z, W). The relational similarity between districts is obtained by computing the Gramian matrix

$$WW' = \begin{bmatrix} 2 & 2 & 1 & 0 \\ 2 & 3 & 2 & 1 \\ 1 & 2 & 3 & 2 \\ 0 & 1 & 2 & 2 \end{bmatrix}$$

Each entry $(WW')_{ij}$ measures the number of contextual regimes shared by districts i and j . Removing the diagonal produces the RelWeights matrix:

$$R = WW' - \text{diag}(WW')$$

$$R = \begin{bmatrix} 0 & 2 & 1 & 0 \\ 2 & 0 & 2 & 1 \\ 1 & 2 & 0 & 2 \\ 0 & 1 & 2 & 0 \end{bmatrix}$$

The degree matrix is obtained from the row sums

$$D_R = \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

The relational Laplacian operator is therefore

$$L_R = D_R - R$$

$$L_R = \begin{bmatrix} 3 & -2 & -1 & 0 \\ -2 & 5 & -2 & -1 \\ -1 & -2 & 5 & -2 \\ 0 & -1 & -2 & 3 \end{bmatrix}$$

The relational Laplacian measures spatial variation across neighborhoods defined by shared contextual regimes rather than purely geometric adjacency.

From a computational perspective, this operator enables several useful analytical extensions. First, the Laplacian defines a graph structure over the relational unit, allowing the application of spectral graph methods such as eigenvector decomposition and spectral clustering. Second, the Gramian matrix WW' can be interpreted as a similarity kernel over spatial units, enabling the construction of kernel operators that capture contextual spatial similarity.

These kernels can be used to propagate information across relational spatial networks. For example, when spatial embeddings are available for one spatial layer (e.g. ZIP-code level socioeconomic embeddings), RelWeights can be used to construct kernel operators that transfer or diffuse these embeddings across intersecting spatial units. This enables embedding propagation across heterogeneous spatial supports.

In practice, such operators allow analysts to move beyond purely geometric notions of spatial proximity and areal interpolation, and instead construct relational spatial operators directly from layered spatial datasets.

Empirical Validation and Implementation

Empirical evaluation of RelWeights was conducted in the context of groundwater scarcity modeling across districts in India Bhogale and Khedgikar (2024) and (Khedgikar, 2023). In the study, districts were intersected with hydrological sub-basins, allowing the authors to extend fundamental resource equilibrium through a spatial analog to study aquifer systems in relation to their administrative boundaries.

Lagrange Multiplier (LM) tests were used to compare the performance of conventional Queen contiguity weights with RelWeights. Across all specifications, the null hypothesis of no spatial dependence was strongly rejected. However, the magnitude of the LM statistics increased substantially when RelWeights were used. In several cases, the LM test statistics increased by approximately five times relative to the equivalent Queen weights specification, indicating stronger detection of spatial dependence when relational spatial structure was incorporated.

An important difference also emerged between model types. Under Queen contiguity, Robust LM Lag tests generally performed better than Robust LM Error tests. In contrast, when RelWeights were used, the Robust LM Error statistics were substantially larger than the lag statistics, suggesting that RelWeights better capture spillovers arising from local externalities embedded within hydrological systems.

The RelWeights approach therefore, revealed spatial dependencies that were not visible under conventional adjacency-based weights. These results suggest that many spatial processes operate through contextual spatial regimes rather than purely geographic proximity.

The RelWeights construction is straightforward to implement. A reference implementation in Python(PySAL) and R (spdep/sfdep) is available at:

<https://github.com/shamilkhedgikar/RelWeights>

The implementation constructs RelWeights using spatial intersection operations and matrix multiplication. The procedure involves generating the intersection matrix between layers, computing the Gramian product, removing diagonal elements, and converting the resulting matrix into a spatial weights object usable in spatial econometric models.

Because the formulation relies only on layer intersections, the approach is easily transferable to a wide range of spatial datasets and modeling environments.

Implications for Data-Driven Spatial Practice

This example illustrates an important principle for contemporary spatial analysis. In many real-world policy settings, spatial relationships are not determined solely by geographic adjacency. Instead, spatial interaction frequently emerges through overlapping contextual systems such as hydrological basins, infrastructure corridors, regulatory regimes, or economic networks.

RelWeights provides a formal mechanism for incorporating these contextual relationships into spatial analysis. By constructing spatial operators from overlay structures, analysts can identify spatial dependencies that remain invisible under conventional contiguity definitions.

In practice, these methods can be applied to problems such as groundwater contamination propagating through shared aquifers, housing market spillovers across transit networks, regulatory effects transmitted through zoning regimes, and environmental exposure across watershed systems.

More broadly, RelWeights illustrates what spatial analysis looks like in a modern data-driven policy environment. Rather than assuming a fixed notion of spatial proximity, analysts can construct spatial operators directly from the layered structure of real-world spatial systems. In this sense, RelWeights provides a bridge between spatial econometrics, planning theory, and the increasingly complex spatial datasets used in contemporary policy analysis (Khedgikar, 2023).

Seminar Outline

The talk building upon this note will cover three main components.

First, the theoretical motivation for RelWeights will be introduced, drawing on ideas from spatial econometrics and urban theory regarding the relationship between geometric space and contextual place.

Second, the construction of RelWeights will be demonstrated through working examples that illustrate how overlay structures generate irregular spatial lags and relational spatial operators.

Finally, the talk will present applications of these methods in real-world planning and policy contexts, including ongoing work within AECOM's Urban Analytics practice, where layered spatial datasets are being combined with increasingly used to model complex urban systems and support data-driven decision-making for DOTs, MPOs and other public sector clients.

Author Bio

Shamil Khedgikar is a Senior Data Scientist at AECOM's Urban Analytics Center of Excellence, where he develops scalable geospatial solutions for public sector clients. With over ten years of experience across academia and industry, his work has focused on research and practice in urban analytics, climate policy, and development economics.

Previously a graduate student affiliated with the Center of Spatial Data Science at the University of Chicago, his work focused on methodological innovations and applications to spatial econometrics. Prior to UChicago, he spent over three years at the Indian School of Business (ISB), first as a researcher in economics and later as a Senior Product Engineer working with the Bharti Institute of Public Policy (BIPP) and the Digital Identity Research Initiative (DIRI). He has also held research positions at the Wharton School UPenn and Cornell University, where he was a graduate student in the College of Architecture, Art and Planning. He also holds a bachelors degree in urban planning from the School of Planning and Architecture Delhi.

His research, supported by organizations such as the Harvard Climate Change Solutions Fund, Bill and Melinda Gates Foundation, and Knight Foundation, focuses on people-centric development and application of spatial tools to inform policies, plans, and priorities.

More Details: <https://shamilkhedgikar.github.io>

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